1739. METHYLOMONAS MEDIUM (SURF)

Solution 1 (20X Basal Salts Solution)

NaCl	31.00	g
Na_2SO_4	3.75	g
KCl	0.52	g
$MgSO_4 \times 7H_2O$	2.46	g
$NaNO_3$	0.85	g
$CaCl_2 \times 2H_2O$	0.88	g

Dissolve the ingredients listed above (in that order) in about 700 ml of distilled water, and then make up to 1 litre

Solution 2 (Wolfe's Vitamin Solution) (see medium 141)

4-Aminobenzoic Acid	5.00	mg
D-Biotin	2.00	mg
Folic Acid	2.00	mg
Pyridoxine-HCl	10.00	mg
Riboflavin	5.00	mg
Thiamine-HCl $\times 2H_2O$	5.00	mg
Nicotinic Acid	5.00	mg
Calcium D-Pantothenate	5.00	mg
Thioctic Acid (Lipoic Acid)	5.00	mg
Vitamin B12	0.10	mg
Distilled water	1000.00	ml

Solution 3 (Acidic Trace Elements Solution) contains the following components per liter of distilled water:

5M HCl	20.00	mL
$FeSO_4 \times 7H_2O$	2.09	g
$ZnSO_4 \times 7H_2O$	0.07	g
$CoCl_2$	0.06	g
$MnCl_2 \times 4H_2O$	0.50	g
$CuSO_4 \times 5H_2O$	0.50	g
$NiCl_2 \times 6H_2O$	0.10	g
H_3BO_3	0.01	g
$La(NO_3)_3 \times 6H_2O$	0.06	g
$Ce(NO_3)_3 \times 6H_2O$	0.06	g
Distilled water	1000.00	ml

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Solution 4 (Alkaline Trace Elements Solution)

NaOH	0.40	g
Na_2SeO_4	0.11	g
Na_2WO_4 x $2H_2O$	0.05	g
Na_2MoO_4 x $2H_2O$	0.29	g
Distilled water	1000.00	ml

Solution #5 (40X Buffer Solution)

HEPES (free acid)	95.30	g
NaOH	11.80	g
Na_2HPO_4	5.68	g
Distilled water	1000.00	ml
Adjust pH 7.8-7.9		

Medium description:
The growth medium contains the following components (per liter):

Solution 1 (20X Basal Salts Solution)	50.0	mL
Solution 2 (Wolfe's Vitamin Solution)	10.0	mL
Solution 3 (Acidic Trace Elements Solution)	0.5	mL
Solution 4 (Alkaline Trace Elements Solution)	0.2	mL
Solution 5 (40X Buffer Solution)	25.0	mL
Distilled water	914.30	ml

Media preparation instructions:

Combine the distilled water and Solutions 1,3,4 in an Erlenmeyer flask and dispense into glass serum bottles.
Flush the headspace of the serum bottles with pure N_2 gas. Seal the serum bottles with butyl rubber stoppers and crimp seals. Using a syringe and needle, inject pure methane and oxygen to final headspace concentrations of 5-10% CH_4 and 21% O_2.
Autoclave the serum bottles for 20 minutes at 121 °C.
Upon cooling to room temperature, add solutions 2 and 5 from a filter-sterilized stock solution.
The final pH of the medium should be 7.8 – 7.9.

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